

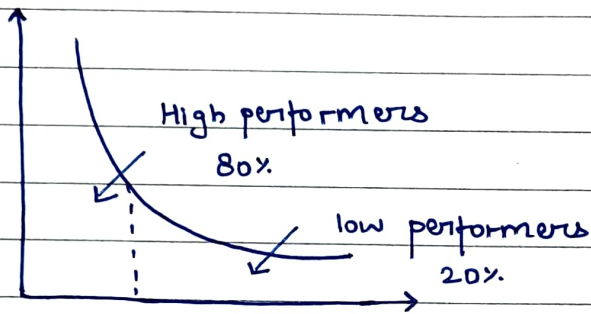
CLASS - 4

Date.....

* Power laws & Scale free Networks

* Pareto principle { 80/20 rule }

→ 80% effect comes from 20% of cause

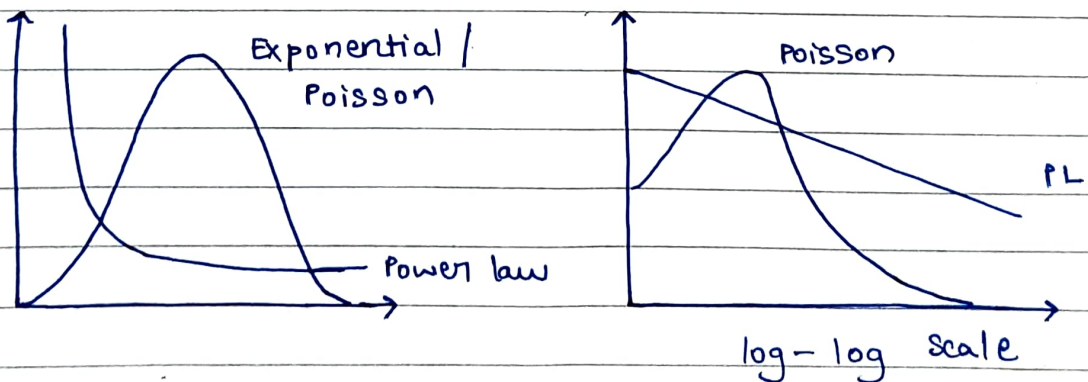


The above follows power law distribution $p(k) \sim k^{-\gamma}$

Most Nodes → low degree

Few Nodes → very high degree (Hubs).

* Power law vs Exponential distribution

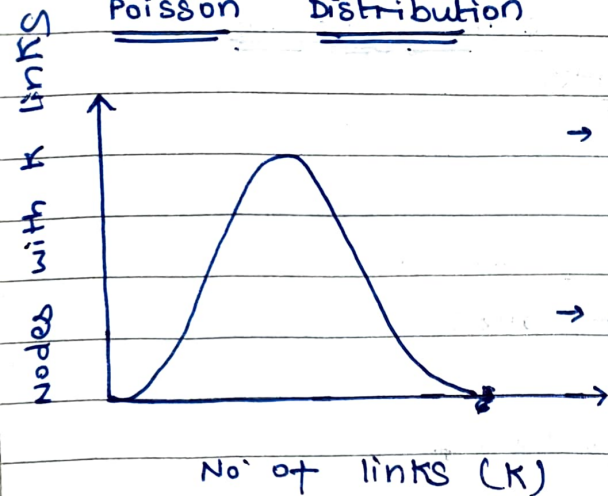


Power law → scale free Networks

Exp / Poisson → Random Networks

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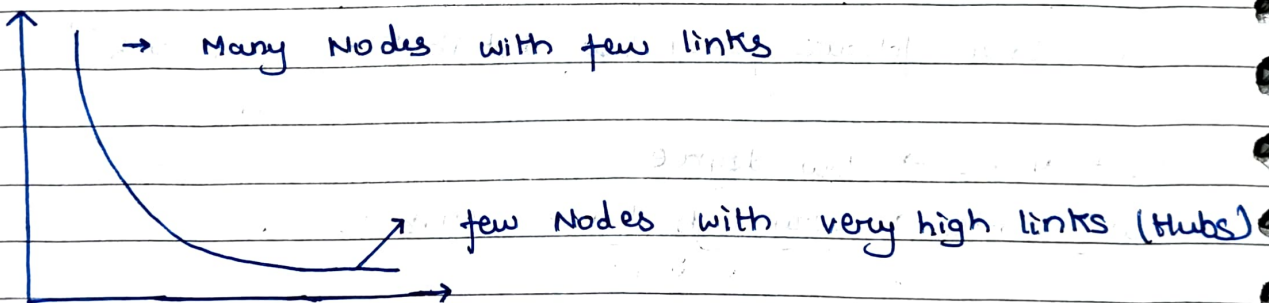
Poisson Distribution



→ Most Nodes have the same number of links

→ No highly connected nodes

Power law Distribution



→ Many Nodes with few links

few Nodes with very high links (Hubs)

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* Finite scale free Networks, size of largest hub

→ All real networks are finite

Expected Maximum degree, k_{max}

$$k_{max} = k_{min} N^{\frac{1}{r-1}}$$

→ k_{max} increases with the size of the network
(As the Network grows, the largest hub also grows)

• 1) when $r > 2$

- k_{max} increases slower than N
- largest hub will contain a decreasing fraction of links as $N \uparrow$

2) when $r = 2$

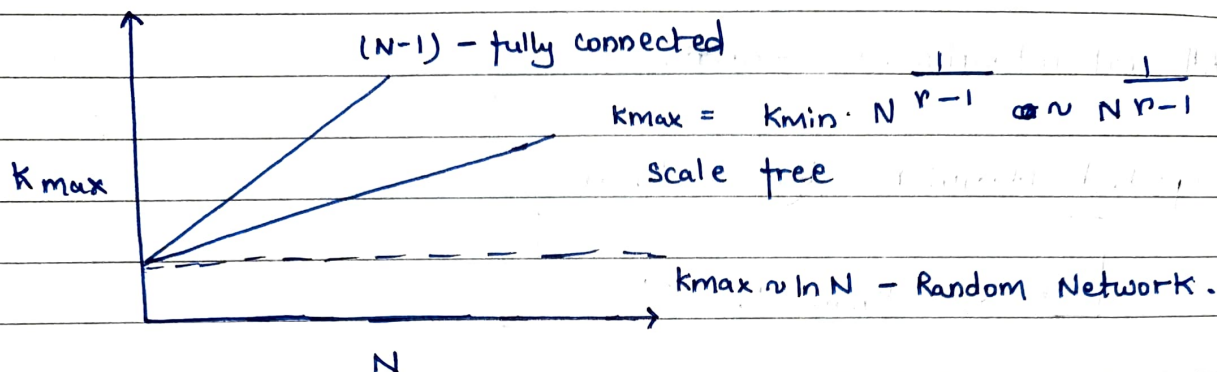
- $k_{max} \sim N$
- The size of the biggest hub is $O(N)$

3) when $r < 2$

- k_{max} increases faster than N { condensation phenomena }
- A single hub captures most links.

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* size of the largest Hub



* scale free Networks

- Networks with a power law tail in their degree distribution are called scale free Networks.
- Degree Distribution → Power law $\{p(k) \sim k^{-r}\}$

* Three Regimes of a Network1) Anomalous Network $\{ r < 2$

→ No large network can exist here

→ $\langle K \rangle$ diverges , $\langle K^2 \rangle$ diverges

→ $\langle d \rangle$ constant , $K_{\max}$ grows faster than N

2) Scale free Regime $\{ 2 < r < 3$

→ Most common regime for real world networks

→ $\langle K \rangle$ finite , $\langle K^2 \rangle$ diverges , $\langle d \rangle \sim \ln \ln N$

→ ~~ego~~ ultra small world

3) Random Regime $\{ r > 3$

→ Indistinguishable from a Random Network.

→ $\langle K \rangle$ finite , $\langle K^2 \rangle$ finite , $\langle d \rangle \sim \frac{\ln N}{\ln \langle K \rangle}$

→ Small world

*

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* No large Network for $r < 2$

for Power law degree distribution:-

$$k_{\max} = k_{\min} N^{\frac{1}{r-1}}$$

If $r < 2$, then $\frac{1}{r-1} > 1$

∴ $k_{\max} \sim N^{>1}$

$k_{\max}$ grows faster than N (Maximum degree grows faster than Network size)

∴ Impossible.

∴ Scale free Networks with $r < 2$ does not exist

∴ Networks with $r < 2$ are not graphical

Configuration Model: a way to build a random network with a given degree sequence.

(a) Assign degrees:

Each node is given a fixed number of "stubs" (half-links),

(b) Randomly connect stubs:

Pick pairs of stubs at random and connect them → this creates edges.

(c) Possible issues:

You might get self-loops (node connects to itself) or multiple edges between same nodes.

(d) Final network:

After pairing all stubs, you get a random network where each node still has the same degree as assigned, but connections are random.

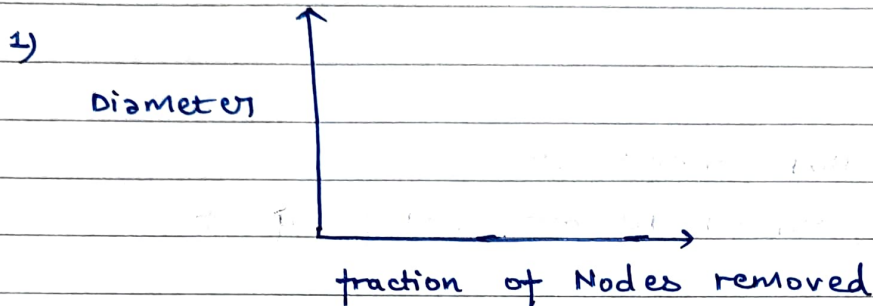
Key idea:

This is basically a degree-preserving random network generator.

★ Scale free Networks are robust to Random failures but ~~fragile~~ vulnerable to targeted Attacks.

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* Response of Random & Scale free Networks to Random Errors & targeted Attacks

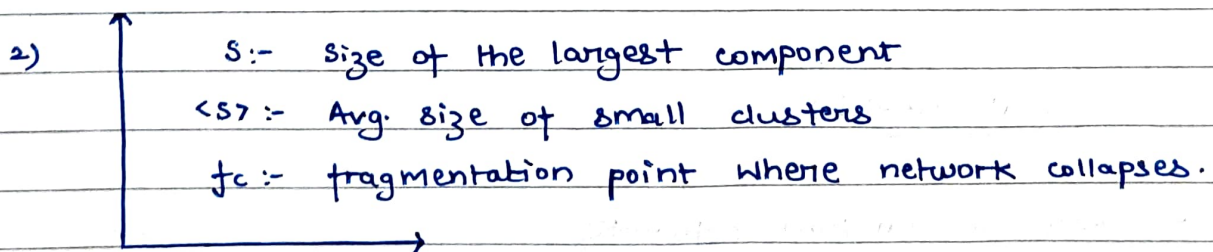


Random Network

- failure & Attack
- Both increase path length gradually as all nodes have similar degree.

scale free Network

- failure - almost No change
- Attack - sharp increase in path length



Random Network → similar curves for both Attack & failure.

Scale free Networks → for failure 'S' decrease slowly ∴ Robust
→ for Attack 'S' decreases quickly ∴ fragile.

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~~original~~

* small world behaviour in scale free

1) $r < 2$

→ $k_{max} \sim O(N)$, one Hub connects all

→ $\langle l \rangle \sim \text{constant}$, Most nodes are 2 steps apart

2) $2 < r < 3$ (ultra-small world)

→ Few hubs dominate

→ $\langle l \rangle = \frac{\ln \ln N}{\ln(r-1)}$, avg path length increases slower than log.

3) $r = 3$

→ some key models have $r = 3$

→ $\langle l \rangle = \frac{\ln N}{\ln \ln N}$

4) $r > 3$ (small world)

→ Almost same as random networks

→ $\langle l \rangle = \ln N$

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* All Networks are not scale-free

A network is not scale free if:-

- 1) degree Dist. is not power law
- 2) Nodes have similar number of connection
- 3) No dominant hubs.

Ex:- 1) Network appearing in material sci - bond b/w atoms of crystalline / Amorphous materials.
→ Each Node has same degree.

2) The Power Grid.

* Distance in Random Graph

$$N_1 \approx \langle K \rangle$$

$$N_2 \approx \langle K \rangle^2$$

$$N_d \approx \langle K \rangle^d$$

$$1 + \sum_{l=1}^{l_{\max}} \langle K \rangle^l = N$$

$$\therefore l_{\max} = \frac{\log N}{\log \langle K \rangle}$$